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Hydraulic Fracturing Evolution in Block 61: TSO, Channel Fracturing and Waterflooding

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Abstract

Block 61 in Ecuador, one of the most productive assets in the Oriente Basin, started a redevelopment phase in 2015 where the introduction of waterflooding in 2017 and hydraulic fracturing campaigns have substantially enhanced oil production. This study starts with an introduction to Block 61 history and reservoir characteristics. A timeline is presented with the highlights in the evolution of hydraulic fracturing technology and waterflooding for the two fracturing campaigns executed since 2015. A comparison of the conventional tip screen out (TSO) and current channel fracturing (CF) is discussed to highlight the differences in fracture design. Formation damage is a continuous challenge in Block 61, that is observed during drilling, completion, workover operations and waterflooding demonstrated by the coreflooding studies. Typical geomechanical properties are outlined by using a type well with fracture jobs in Lower U Sandstone and Lower T Sandstone. This section includes recommendations on channel fracturing candidate selection based on Young's modulus and closure pressure. With the advances in channel fracturing significant improvements were achieved like a reduction in early screenout from 24% to 0% along with the KPIs that demonstrate the overall success of the project. Examples for wells fractured in waterflooding influenced areas are provided to understand the production scenarios where both technologies are combined. A summary of hydraulic fracturing candidate selection for wells in waterflooded influenced areas is included. Finally, the project is evaluated using three key performance indicators, oil production before and after fracturing, inflow performance relationship and historical oil production after capex intervention. All key performance indicators show not only productivity increase, but it can be sustained over time with an according waterflooding and reservoir management strategy. Hydraulic fracturing in Block 61 resulted in more than 7,000 stb/d (approximately 10% of the total oilfield production comes from the nine wells presented in this study) and other highlights like the development of a new horizon, Upper T Sandstone, and the two biggest fracture jobs in Ecuador so far.

Introduction

Block 61, one of the most productive assets in Ecuador, composed of 17 fields, is located central play of the Oriente Basin (Sacha – Shushufindi corridor). Limits to the north with Sacha field, to the South with Block 17; to the East with Blocks 14 and 17 and to the west with the Puma and Pucuna fields, [Fig. 1](#). The block extends 43 miles from north to south with an area of 783 km² with an estimated STOIP of 3.9 billion STB. The main reservoirs are highly heterogeneous Cretaceous fluvio-estuarine (Hollin, Lower T and Lower U sandstones) and secondary reservoirs are shallow marine sandstones (Upper Hollin, Upper T and Upper U sandstones). It was discovered in 1970 and commercial production started in 1975 with additional exploration and appraisal drilling campaigns in 1980, 1991, 2009, 2013 and 2016. In December 2015 a new exploitation strategy started with the goal of tackling production decline, 31% per year and accelerating reservoir pressure depletion, thereby implementing secondary recovery, infill drilling and well interventions such as reperforating, acid stimulation, hydraulic fracturing and others as reported by [Morales, 2017](#), [Fun-Sang, 2018](#), [Mendoza, 2019](#), [Mata, 2023](#).

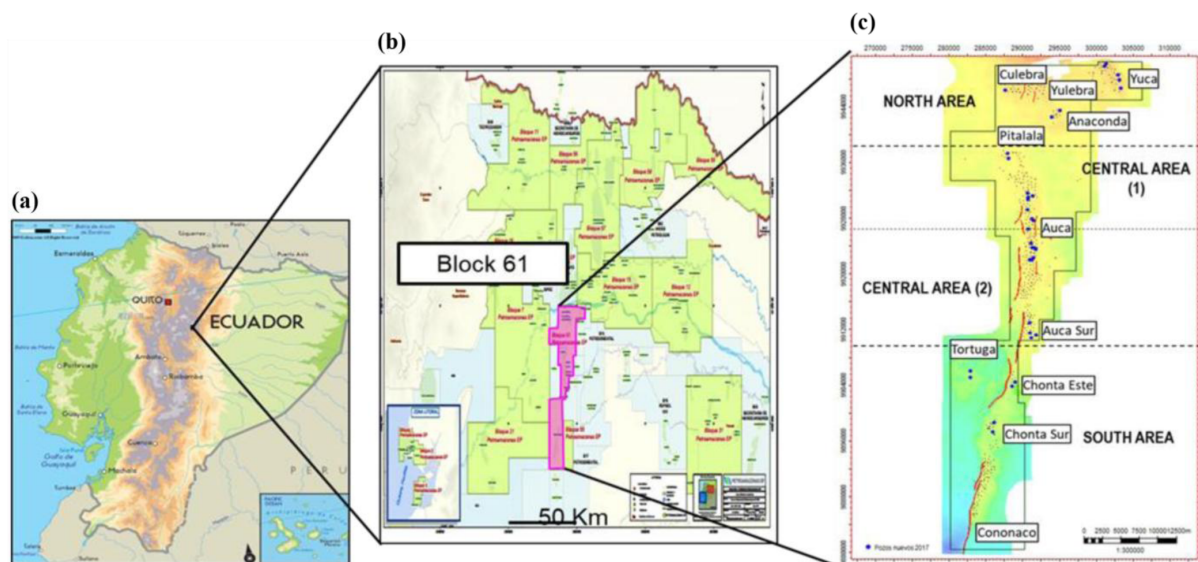


Figure 1—Block 61 Location: (a) Geographical location in Ecuador; (b) Block 61 highlighted in purple with offset blocks and fields; (c) Fields that compose Block 61, Operation areas and water injection wells depicted in blue

The low salinity waterflooding project (peripheral and pattern) started in 2017 in four different reservoirs (Basal Tena, Upper U, Lower U and Lower T) with six pilots located in areas with reservoir pressure near to bubble point pressure, good reservoir continuity and available drainage points. In 2018, initial results showed incremental oil production between two and three months after injection started and positive reservoir response. These results proved the benefits of water injection promoting new drilling campaigns, additional waterflooding patterns and well interventions to restore productivity of damaged wells in influenced areas. ([Mendoza, 2019](#); [Piñeiros, 2020](#)). In 2020, waterflooding was expanded and accelerated along with production/injection pattern balancing, artificial lift optimization, water management, reservoir surveillance, and new completion technology. As of 2025, 53 irregular injection patterns are in operation in five reservoirs (Basal Tena, Upper U, Lower U, Upper T, Lower T) with a total production of 72,739 stb/d, with 64.59% water cut.

Hydraulic Fracturing Timeline

The use of hydraulic fracturing (tip screen out, TSO and channel fracturing, CF) has significantly restored the productivity of conventional high and medium permeability wells of Ecuadorian oil fields. [Izurieta,](#)

2022, presents a literature review from 2013 to 2022 including the technique, field and reservoir in the Oriente basin. More recent studies report production results and guidelines for a successful fracturing campaign (Salazar, 2023) and the use of non-degradable fibers combined with resin coated proppant, RCP, for proppant flowback control (Salazar, 2024) both studies in an offset fields to Block 61.

Hydraulic fracturing has been a key operation to increase oil production along with waterflooding in Block 61 since 2015 as part of the new development strategy. Fun-Sang, 2018, reported that 30% of the incremental production was related to formation damage removal jobs such as reperforating, acid stimulation and hydraulic fracturing and Piñeiros, 2020, 30%, related to waterflooding demonstrating the combined impact to maintain production of both technologies. Fig. 2, shows a timeline with the most relevant events for hydraulic fracturing and waterflooding from 2016 to present including some recent milestones and Fig. 3, describes the hydraulic fracturing activity for both fracturing campaigns along with the reservoirs stimulated from 2016 to present.

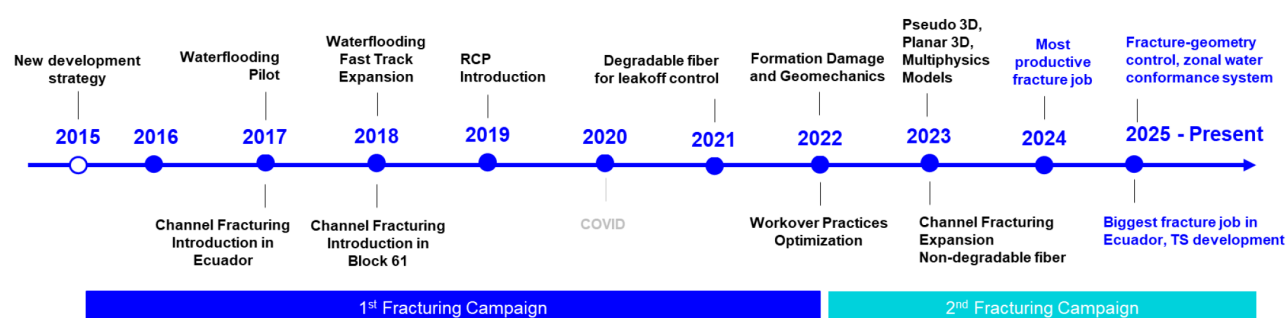


Figure 2—Timeline describing the main events since 2015 for waterflooding and hydraulic fracturing in Block 61, recent milestones are highlighted in blue

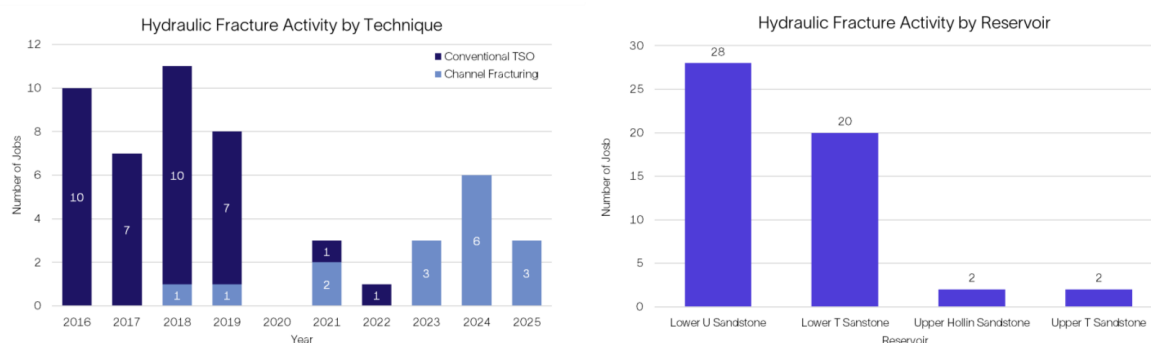


Figure 3—Left, Historical hydraulic fracturing activity by year. The first fracturing campaign started from 2016 to 2022. The second fracturing campaign started in 2023 and continues to be executed at the time of this study. Right historical hydraulic fracturing activity by reservoir. 92% of jobs are focused on the Lower U and Lower T sandstones where the waterflooding is also focused on

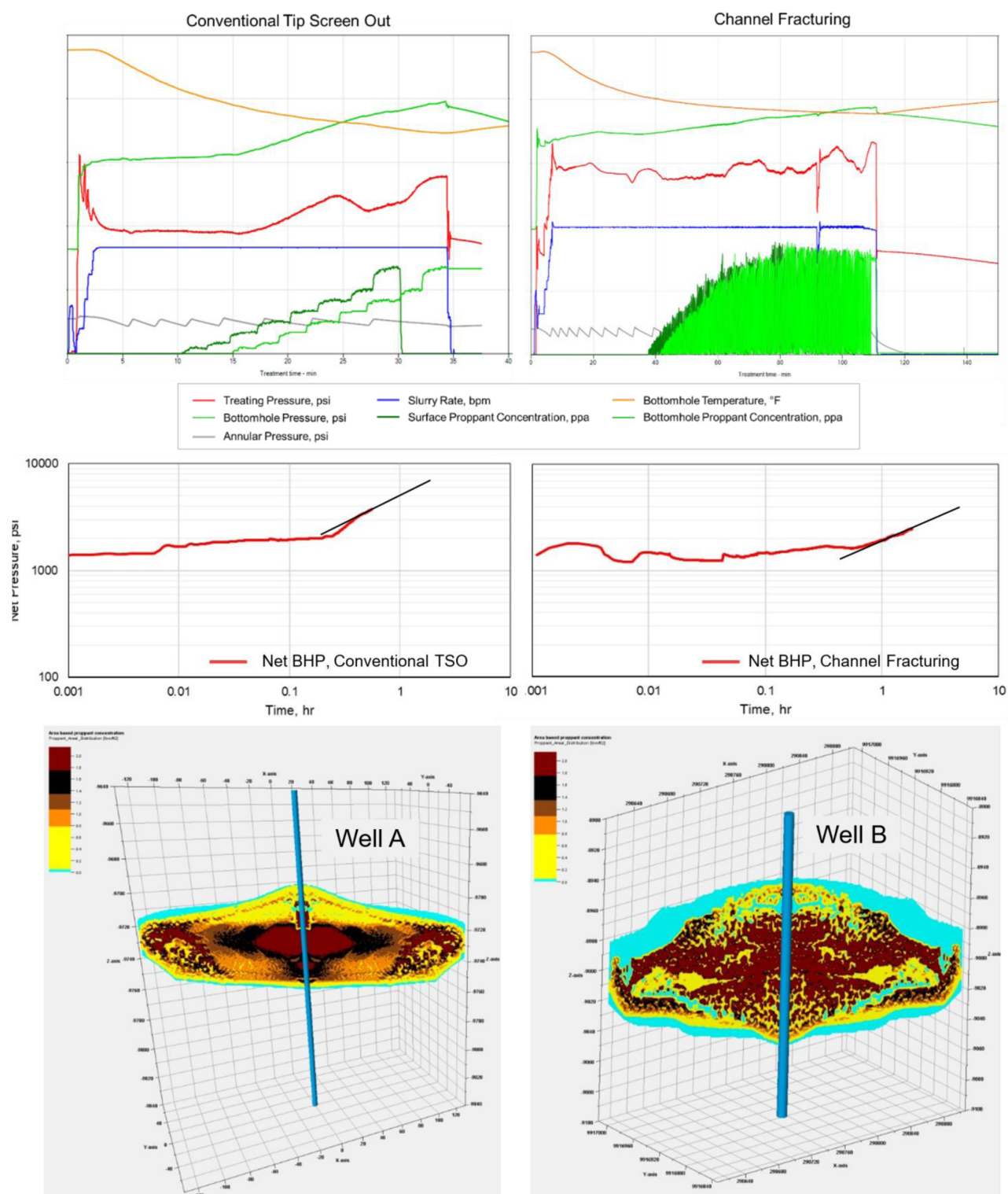
The first fracturing campaign started in 2016 using conventional TSO. These jobs use a chelating agent/viscous acid pre-pad that requires soaking time to dissolve scale and fines in the near wellbore region and perforations. Next, a pre-frac injection test is pumped using the same gel as the main job to calibrate the fracture model and perform any modification in the pumping schedule if required. The main job consisted of an average of 28,000 lb LDP ceramic proppant, 20/40 mesh, pumped in stages from 1.0 ppa to 7.0 ppa using crosslinked borate-based gel along with fiber-based proppant flowback control (FBPFC) additive (Ramones, 2014). Finally, the treatment was displaced to top perforations at the maximum proppant concentration.

This stage faced some challenges leading to mixed production results because of a combination of factors, such as low reservoir pressure early screenout, 24% of jobs, proppant flowback, and sometimes skin damage post frac. To improve success rate, a degradable fiber is mixed with the gel to reduce the leakoff but also

to limit the polymer invasion in the reservoir and hence polymer damage and for proppant flowback resin coated proppant (RCP) was pumped in the last proppant laden stage.

In 2021 to 2022 several operational changes were implemented to optimize workover practices, leading to reduced stress cycling and formation damage risk. Channel Fracturing was introduced in Ecuador in 2017 and in 2018 in Block 61, but it was not fully adopted until 2023 during the second fracturing campaign. In Block 61, 18 jobs have been executed and 10 of them in the second fracturing campaign, from 2023 to present. Currently the incremental oil production derived from these jobs exceeds 7,000 stb/d and 5,0 MMbbls cumulative oil production. CF is the standard technique used because the low screenout rate and positive production results, currently more than 96% of the fracture treatments in Ecuador use this technique (Salazar, 2024).

Current operations use a modified pre-frac injection test that includes fiber for leakoff control, and two sand slugs to reduce perforation and tortuosity friction. These slugs are over flushed with crosslinked and linear gel. The main job uses an average of 59,000 lbs LDP ceramic proppant, 16/20 mesh with proppant concentration ramped to the maximum concentration, 10.0 ppa, using high frequency pulses with most cycles at maximum concentration. The final proppant laden stage, Tail-in, is a non-pulsing stage, uses RCP and non-degradable fiber, that increases the frictional forces chemical resin bonded, to create a high conductivity flexible proppant pack, Salazar, 2024. Then the treatment is displaced with linear gel to leave up to 500 lb in the wellbore. For these jobs, up to 3 types of fibers (degradable and non-degradable) and two proppant types (RCP and non-RCP) can be combined according to the reservoir characteristics. Since most fracture jobs are executed in waterflooding affected areas, Lower U and Lower T sandstones, some of the jobs include a relative permeability modifier, RPM, pre-pad. Fig. 4 shows operation charts, fracture geometry and proppant distribution for typical TSO and CF to highlight the design differences described previously. Latest fracture jobs target a new horizon, the Upper T Sandstone (TS) in areas where it develops a low permeability, layered reservoir and production without fracturing is not economical. For these jobs the proppant mass has increased compared to traditional LU and LT resulting in a new reservoir to be developed and the two biggest fracture jobs in Ecuador so far.



Reservoir properties and Formation Damage

The Lower U (LU) and Lower T (LT) sandstones hold 96% of fracture jobs and are the most successful in waterflooding results. These sandstones are members of the Cretaceous Napo formation; both are complex stratigraphic reservoirs that exhibit drastic thickness changes as well as lateral and vertical heterogeneity. They present good petrophysical properties, porosity (from 13% to 18%) permeability (from 200 to 750 mD), oil from 15 to 32° API, associated with fluvial and estuarine facies (tide dominated estuaries that grade to marine deposits) with fining upward sequences. As of 2015, both reservoirs the pressure decreased more than to 40% of the original pressure, reaching values close to the bubble point pressure, resulting in low production rates and low productivity negatively impacting the value of the asset economics and reflecting in a low recovery factor (Mendoza, 2019, Piñeiros, 2020). The low productivity from 2015 to 2018 was also a result of formation damage related to fluid invasion during drilling, completion and workover operations, fines migration and scale precipitation derived from water breakthrough, Fun-Sang, 2018. Table 1 shows composition for some core samples of these reservoirs composed of quarts and clays such as kaolinite and illite and chlorite, migrating fines and smectite, a swelling clay.

Table 1—XRD results for selected samples from the Lower U and Lower T cores. The main mineral in the samples is quartz with clays such as kaolinite and illite, migrating fines and smectite a swelling clay, modified from Fun-Sang, 2018.

Well	Sand	Quartz	Kaolinite	Carbonate	Pyrite	Feldspar	Illite/ Glauconite	Other	Chlorite	Smectite
CLBA-006	LU	99%	-	-	-	1%	-	-	-	-
ACAB-064	LU	95%	2%	-	1%	1%	-	-	-	-
YCA-013	LU	99%	1%	-	-	-	-	-	-	-
YLB-007	LU	95%	1%	-	-	-	-	-	-	4%
YLB-007	LU	97%	2%	-	-	-	-	1%	-	-
ACAB-16	LU	93%	2%	-	-	1%	-	-	-	4%
YCA-002	LT	59%	-	5%	1%	-	32%	3%	-	-
YCA-002	LT	71%	-	7%	-	-	13%	-	9%	-
ACA-030	LT	91%	-	7%	-	1%	-	1%	-	-
ACAO-037	LT	90%	1%	-	-	-	6%	-	-	4%
YCA-013	LT	88%	1%	-	-	-	7%	-	-	5%
ACAB-064	LT	90%	1%	-	-	3%	-	-	-	6%
ACAC-080	LT	98%	1%	-	-	1%	-	-	-	1%

During the second fracturing campaign, a formation damage study was conducted to understand the effects of the completion brine and the water injection on the stimulated wells. The injection water is produced from the Main Hollin sandstone aquifer has an average salinity of 600 ppm Cl⁻. The same water is used for the completion brines for workover operations and fracturing fluids. Typical completion fluid composition includes potassium chloride, mutual solvent, surfactant, bactericide and clay control additive if necessary. Results from this project demonstrated that formation damage could arise from completion brines as shown in Table 2 as well as from waterflooding, Table 3. A reduction in productivity is observed in the three samples as the water saturation changes to 100% formation water to 100% injection water. Sharma, 2007, describes this phenomenon. As salinity reduces, so do the forces that hold fines on the pore surfaces (electrostatic and van der Waals). Once fines migrate, pore throats plug, and flow area is reduced (i.e. formation damage). Based on these results, higher fracture conductivity was promoted for the current fracture campaign as previously described.

Table 2—Relative permeability before and after core flood with completion brine.
All tests are conducted at net overburden pressure and reservoir temperature.

Sample	Porosity	Absolute Permeability	Before completion brine				After completion brine				Production Loss
			S_{wirr}	$K_{eff\ oil}$	S_{or}	$K_{eff\ water}$	S_{wirr}	$K_{eff\ oil}$	S_{or}	$K_{eff\ water}$	
	%	mD	%	mD	%	mD	%	mD	%	mD	
A	12.8	180	30.9	77.2	41.2	4.30	30.9	17.2	45.8	0.701	-17%
B	16.3	560	28.0	174	42.3	15.7	28.0	102	50.3	7.12	-27%
C	16.4	2,108	17.2	1,329	43.4	63.5	17.2	874	48.3	39.7	-12%

Table 3—Absolute permeability to water for different formation water (FW) and water injection (IW) mixture, all experiments conducted at net overburden pressure and reservoir temperature. The final test returns to 100% FW to simulate a workover operation confirming the formation damage when the rock is exposed to any low salinity fluid.

Sample	Absolute permeability to water (mD) for Formation Water (FW) / Injection Water (IW) Mixture					
	100% FW	75%FW / 25% IW	50%FW + 50% IW	25% FW + 75% IW	100% IW	100% FW
AA	35.3	28.9	26.5	25.1	21.2	26.4
BB	190	92.2	31.3	19.5	16.7	19.7
CC	95.6	66.9	23.7	19.5	15.9	22.1

Geomechanics

Reservoirs in Block 61 are in normal stress regime with minimum horizontal stress (closure pressure, $p_{closure}$), S_{hmin} from 0.442 to 0.777 psi/ft, overburden gradient of 1.03 psi/ft, Fig. 5, and maximum stress is assumed to have 300 psi difference from S_{hmin} and direction in the range of 110° to 120°, following the regional trend as reported by Torres, 2010. Fig. 6, shows a type well with sonic dipole logs and associated pre-frac injection test in both LU and LT sandstones. In this example, dynamic properties, Young's modulus (YM) and Poisson's ratio (PR), are calculated directly from sonic and density logs and converted to static using correlations (Barree, 2009). Geomechanical properties estimation is key for the selection of channel fracturing. A low YM to closure pressure ratio, $YM/p_{closure}$, could signify the formation to collapse on the channels creating pinch points that could impact fracture conductivity. A $YM/p_{closure}$ ratio above 350 is recommended for CF applications with a minimum of 200 reported by Ajayi, 2011. Other guidelines recommend CF to be used on $YM > 2.5 \times 10^6$ but successful applications have been reported in YM as low as 1.0×10^6 (Azhari, 2017). Both fracture jobs were executed with the channel fracturing technique in different fracture campaigns.

- The LT formation was fractured in the first campaign, 2021. It is a massive sandstone, medium permeability, 83 md, with YM of 4.14×10^6 psi, PR of 0.18. The pre-frac injection test was conducted at 17 bpm using 30 lb/Mgal borate-based. The estimated $YM/p_{closure}$ of 930.
- The LU formation was fractured in the second campaign, 2025. It is a high permeability sandstone, 1,258 md, the well is part of the Pad J waterflooding pattern in the central area of Block 61. YM of 3.74×10^6 psi, PR of 0.26. The pre-frac injection test was conducted at 20 bpm using 45 lb/Mgal borate-based gel. The estimated $YM/p_{closure}$ of 689.

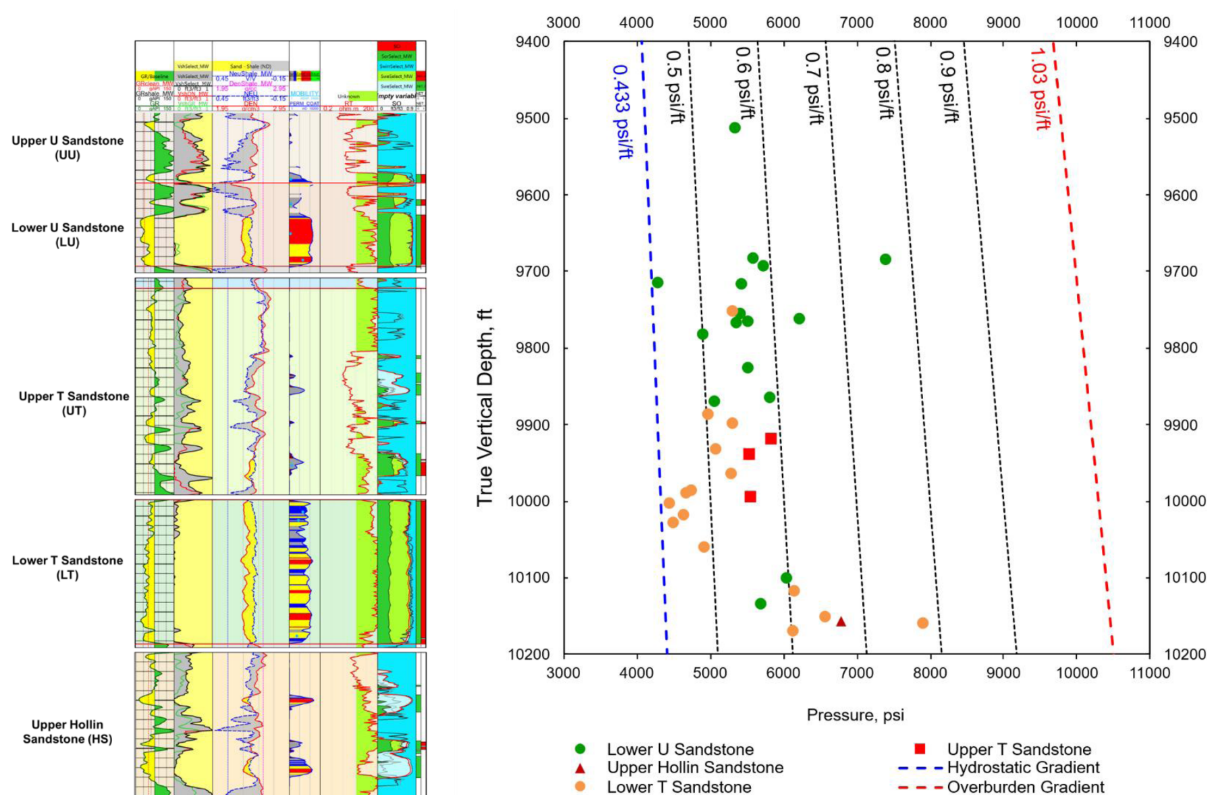


Figure 5—Typical Well log (Left) and Closure pressure for the wells fractured in Block 61 since 2015 to present by reservoir (Right). Gradient lines are plotted as reference

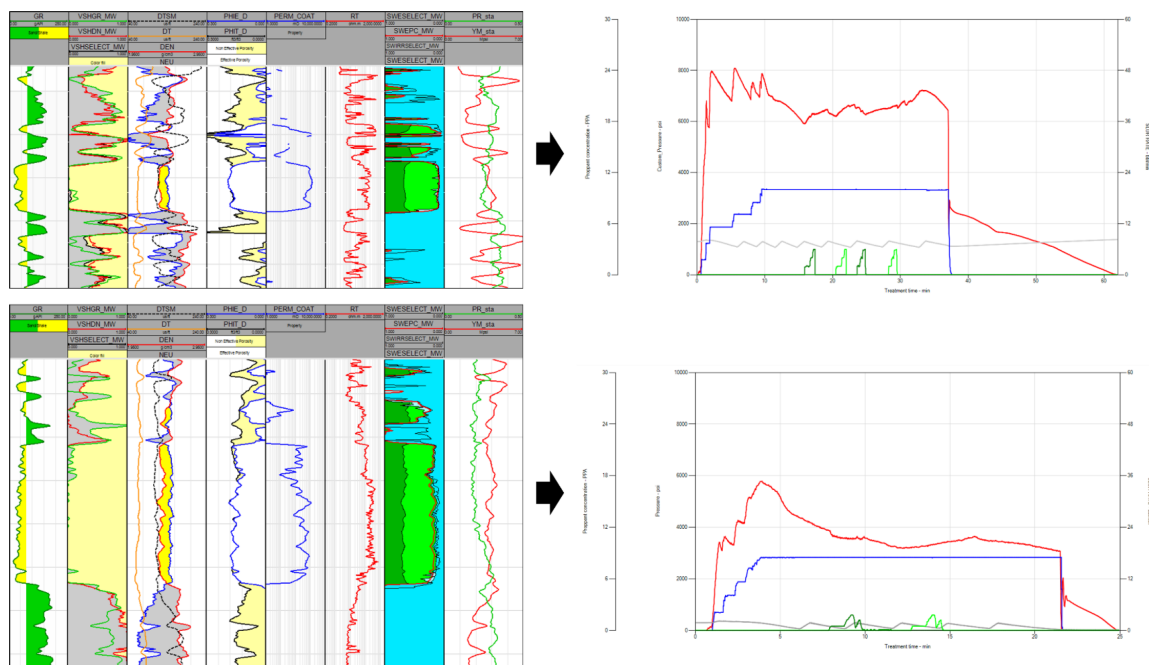


Figure 6—Well logs for the Lower U Sandstone (Top left) and Lower T Sandstone (Bottom left). Track 1, Gamma ray highlighting cutoff for sand in yellow and green. Track 2, Shale volume from gamma ray, neutron & density, Track 3, Porosity logs, neutron, density, compressional and shear transient time, Track 4, Absolute permeability, Track 5, Deep Resistivity, Track 5, Water saturation, irreducible water saturation, mobile water saturation, Track 6, Static Young's modulus and Poisson's ratio, dynamic properties are calculated directly from logs and converted to static using correlations because the limited core data. On the right is the corresponding pre-frac injection test. The Top right executed on the second fracture campaign and the bottom right during the first campaign

Hydraulic Fracturing, Waterflooding and Production Strategy

Water injection patterns in Block 61 have irregular shapes. As the development continues, new drainage and injection points are open to increase overall sweep efficiency. Hydraulic fracturing not only enhances well productivity but also contributes to improving vertical and areal sweep efficiency. Fracture jobs in waterflooded areas show high productivity performance with two general cases. All cases show immediate production increase after fracture stimulation; other wells show a second production peak before water breakthrough in the fractured well, Fig. 7 and Fig. 8 shows different production profiles and examples of patterns and fractured well locations inside and outside water injection patterns.

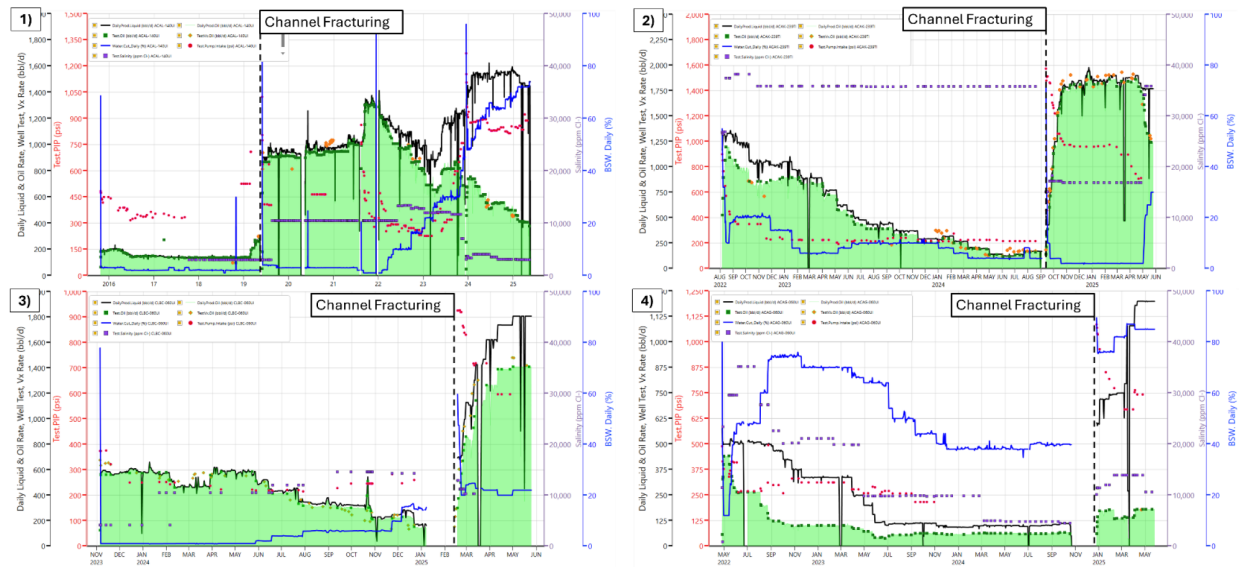


Figure 7—Production performance after hydraulic fracturing in waterflooding influenced areas. Cases 1) and 2) show incremental production after fracturing and later water breakthrough. Cases 3) and 4) immediate production increase after fracture stimulation with stable water cut

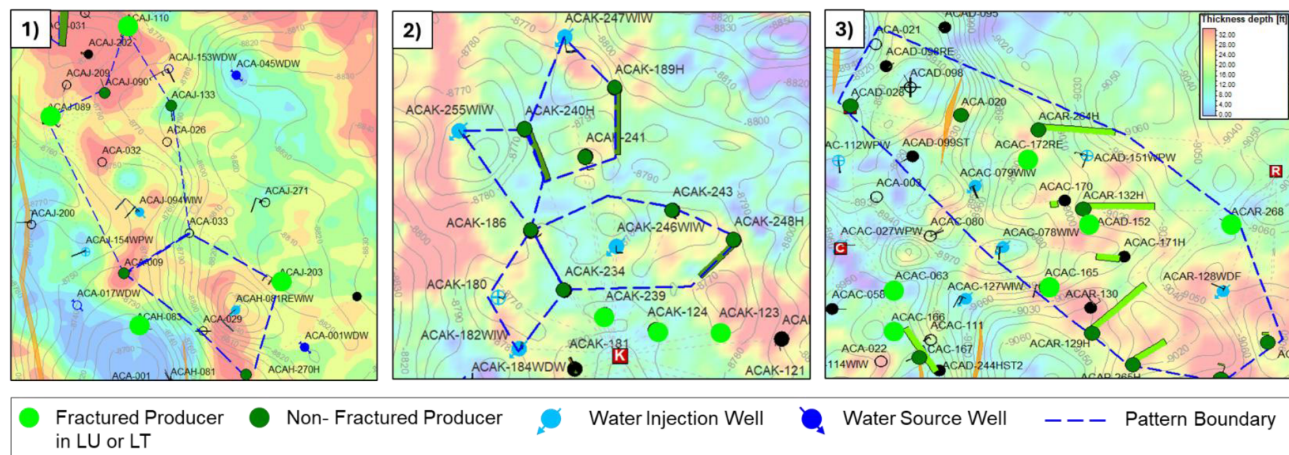


Figure 8—Examples of waterflooding patterns in Block 61. 1) Fractured wells are part of the pattern to improve sweep efficiency to drainage points with limited effect. 2) Fractured wells located in the south area of the pattern promote better connectivity in unsweep areas and to manage water production already influenced areas. 3) Massive pattern that combines various injectors, horizontal and fractured wells, horizontal laterals are highlighted in green.

Fig. 7, Case 1) shows an example of a well fractured in mid-2019, before the waterfront reaches the producer; after two years with no significant water production, a second peak oil is observed then the water breakthrough. In this later stage, fluid production increases and the pattern is balanced to control water cut. Case 2) This example shows a well with high formation damage in an area where the reservoir pressure has

increased because of the ongoing waterflooding. After fracturing, a significant incremental oil production is observed with stable oil production for about eight months with no decline as result of pattern balancing. Recently, the water cut started to increase as result of water breakthrough. Case 3) A well fractured in a mature pattern with excellent results in both productivity and water cut. This job included the use of RPMs to delay water production with a positive response. Case 4) Similar to previous case but significant water cut increase post fracturing. In this case the pattern balance actions had limited effect to control water cut post fracturing. To balance patterns in Block 61 different tools are used to monitor and estimate allocation factors related to each injector, such as reservoir simulation, streamlines and capacitance resistance model (CRM) combined with machine learning and analytical modeling, [Paredes, 2023](#), shows a recent application of these tools in a Block 61 oil field.

Candidate Selection

For the second fracturing campaign some guidelines were adopted to locate low risk candidates in waterflooding influenced areas, [Table 4](#) summarizes these recommendations. Candidates with high skin and permeability are the best performers. This is either defined by pressure transient analysis (PTA) or rate transient analysis (RTA). These cases pose a challenge because of the high leakoff and the risk of fracture conductivity could not overcome reservoir conductivity. For these scenarios, the modifications in fracture design had significantly increased the chance of success by reducing early screenout ratio from 24% to 0% and by using 16/20 LDP ceramic proppant at higher proppant concentration have secured a highly conductive fracture. Candidates in low reservoir pressure areas are the most damaged during drilling and completion operations. For these cases, the best strategy is to identify skin from PTA, define productivity scenarios and be fractured once an injector has pressurized the area or there is evidence of oil bank displacement in nearby wells.

Table 4—Candidate Selection guidelines for the second fracturing campaign under local project economic targets for LU and LT Sandstones

Category	Condition
Mechanical Integrity	• Good cement quality to avoid behind casing communication with unwanted zones • Well integrity evaluation to ensure that tubular can handle treating and annular pressure • Rathole, > 50 ft to avoid wellbore cleanup operations
Petrophysics	• High reservoir thickness, > 30 ft • High absolute permeability, > 200 md • Zones with low current water saturation: require reservoir simulation or PNL/CO evaluation and fractional curve analysis • No transition zone or water oil contact present
Productivity	• Recent pressure build up (PTA) with positive skin • Analytical model (RTA) with matching reservoir and flowing pressure that confirms positive skin • Fast fluid production decline, not associated with reservoir depletion • Complete nodal analysis including available ESP and surface limitations
Reservoir	• Reservoir pressure above 1,000 psi • Waterflooding influenced areas, water cut < 30% • If water breakthrough is observed in offset wells, detailed numerical reservoir simulation • Water injection capacity for pattern balancing, VRR > 1.2 or 1.0

Results

To evaluate the performance three key performance indicators were defined 1) production before and after fracturing 2) inflow Performance Relationship and 3) historical production after capex intervention, this profile includes total well production for wells in commingled production where LU or LT was hydraulically fractured (Mejia, 2024). Fig. 9, shows the before and incremental production post fracturing for the 2024 – 2025 campaign with overall positive results. Fig. 10, shows inflow performance relationship (IPR) before and after for wells fractured in LU or LT and post frac IPR only for UT. As noted, all the cases show a positive improvement in productivity index, demonstrating a proper candidate. Finally, to evaluate the performance over time, Fig. 11, shows the post frac production that reaches 7,000 stb/d approximately as of May 2025.

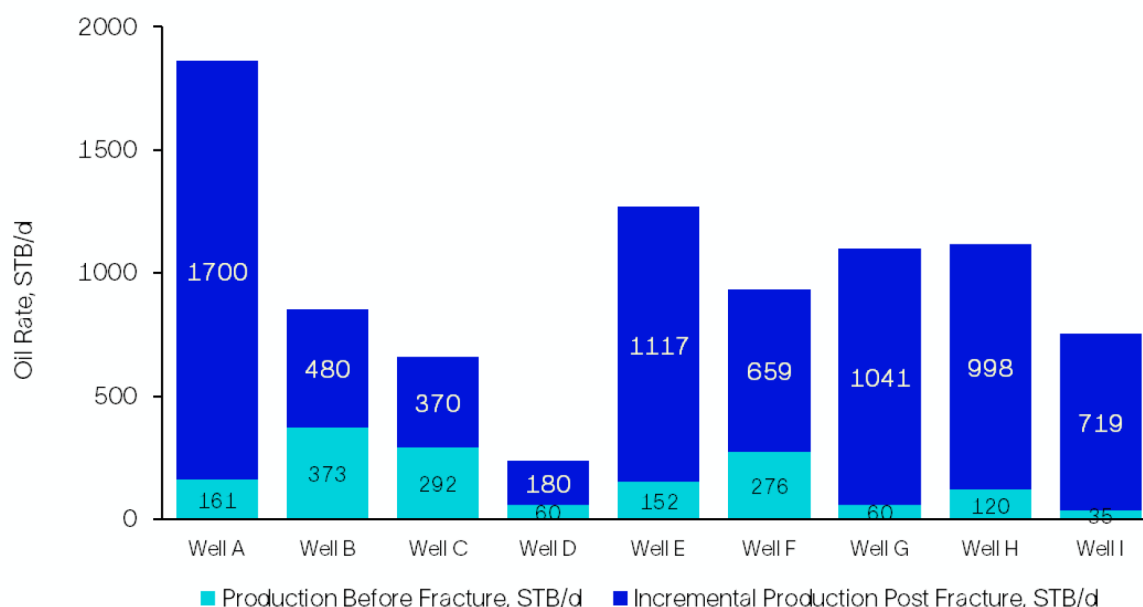


Figure 9—Before and incremental production post fracturing for the 2024 – 2025 campaign

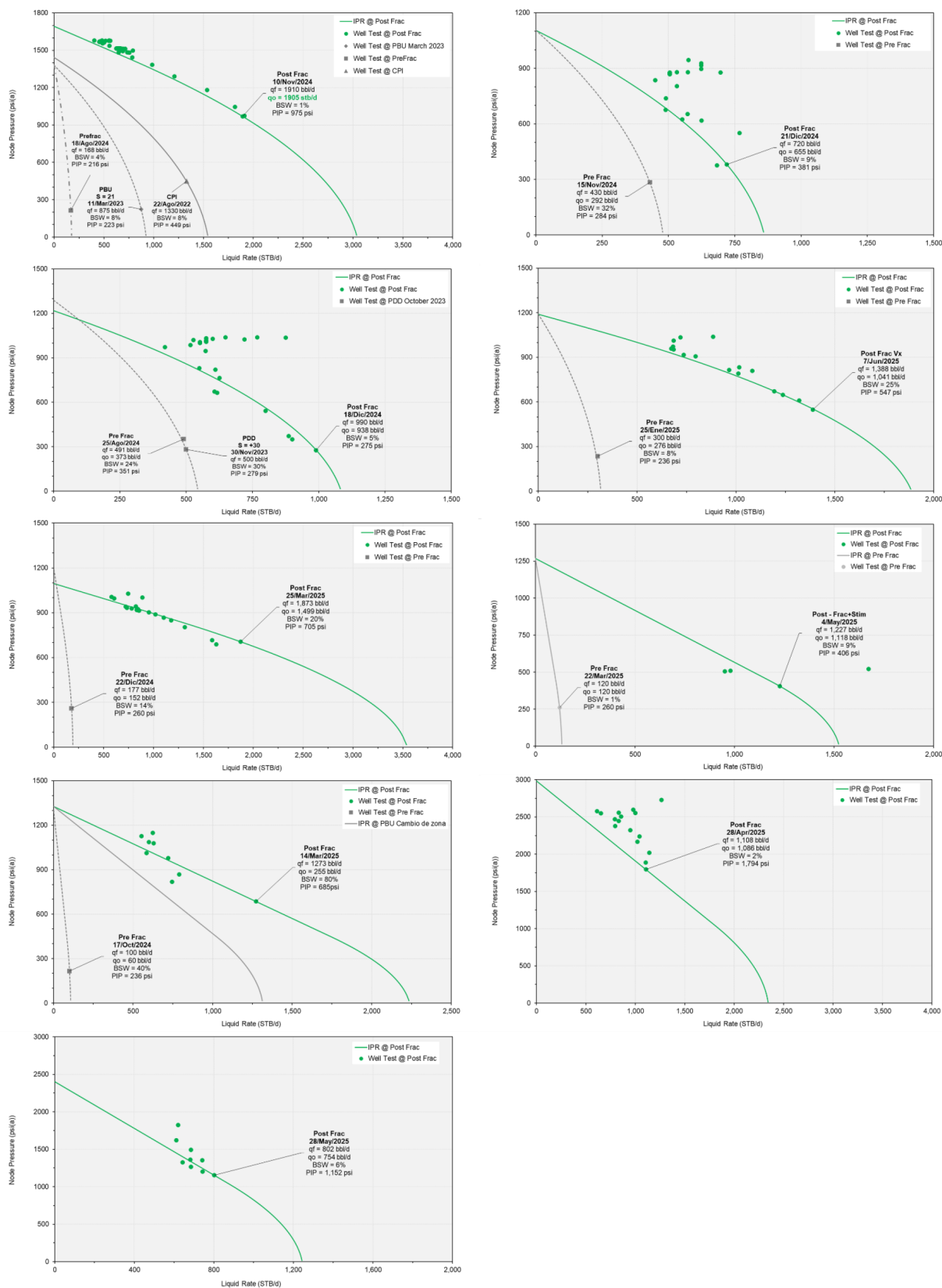


Figure 10—Inflow Performance Relationship before and after for wells fractured in LU or LT and post frac IPR only for TS, for the 2024 – 2025 campaign

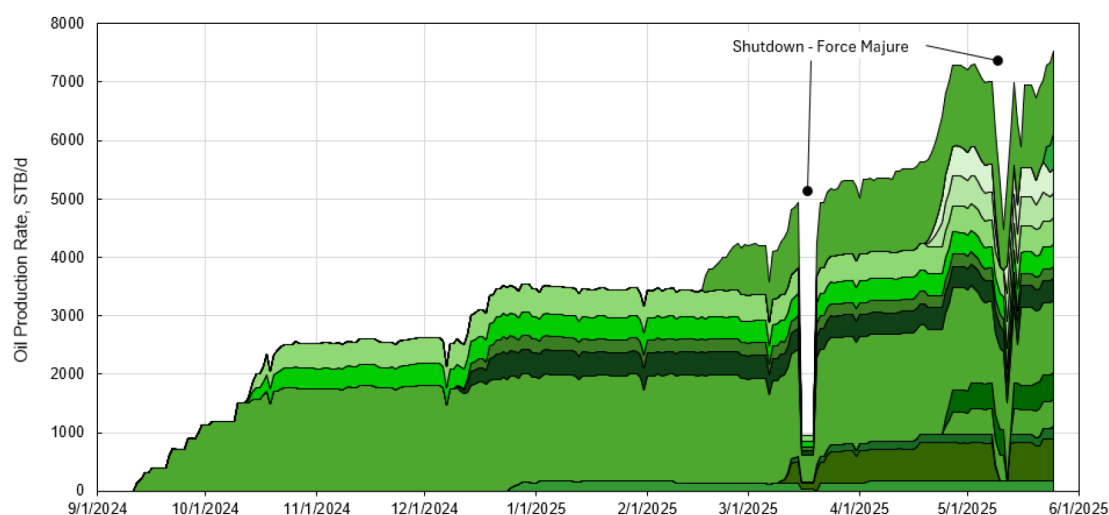


Figure 11—Historical production after capex intervention, this profile includes production from fracture and commingle production

Conclusions and Recommendations

Results demonstrate not only a significant increase in productivity but also the potential to sustain it over time with an appropriate waterflooding and reservoir management strategy.

Hydraulic fracturing in Block 61 has delivered outstanding results, contributing more than 7,000 stb/d, approximately 10% of the total oil production of the field. This success includes the development of a new horizon, the UT sandstone, and the execution of the two largest fracture jobs in Ecuador to date.

The integration of expertise across all disciplines was critical to addressing challenges encountered during the first fracturing campaign. This collaborative approach led to key modifications that yielded exceptional results, including the significant production gains and the successful development of new horizons.

The introduction of channel fracturing, combined with a deep understanding of reservoir characteristics and effective candidate selection, has resulted in the most successful fracturing campaign in Block 61. These best practices can be adapted and applied to other fields facing similar challenges, offering a roadmap for future success.

The waterflooding project in Block 61 stands as the most successful in Ecuador. By documenting and sharing the specific challenges and solutions associated with hydraulic fracturing, this study provides valuable insights that will enhance future implementations and foster knowledge sharing within the technical community.

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